

## Unit 5: Make It, Mend It, Wear It

**NYS content module:** Apparel and Textile Production and Design (ATPD) **Length:** 28 class periods of 40 minutes (days 116 to 143 of the year; days 116 to 135 sit in marking period 3, days 136 to 143 in marking period 4) **CTE themes carried:** Sustainability (primary); Financial and Consumer Literacy; Problem Solving and Innovation; Health, Safety, and Wellness; Communication and Interpersonal Relationships; Career and Community Opportunities **Unit project:** Upcycle Challenge, scored with Rubric 05 - Upcycle Challenge.md **Companion files in this folder:** Calendar 05.md, Vocabulary 05.md, Lessons 5.1 to 5.19, Project 05 - Upcycle Challenge.md, Rubric 05 - Upcycle Challenge.md, Assessment 05.1 - Fibers and Care Quiz.md (given inside Lesson 5.4 Day 2), Assessment 05.2 - Stitch Sampler Performance Check.md (given inside Lesson 5.11 Day 3), Assessment 05.3 - Unit 5 Test.md (given inside Lesson 5.19 Day 2). The hand-sewn item rubric lives inside Lesson 05.10. Handouts and slide outlines are in 05 Materials/Unit 05/.

Reconciliation note: this plan was written to the scope and sequence and then reconciled to all nineteen lessons, the project, the rubric, and all three assessments as they landed in the folder. Where a lesson page and this plan disagree, the lesson page wins; the calendar and this plan follow the lessons' day counts, handout names, and check slots.

### 1. Why this unit

I took a graduate course called The Meanings of Dress, and the thing I wrote about was dress as communication and identity: that clothing says something to other people whether or not you meant it to. Then Spirit Week happened. After 80s Day the hallway comments got mean, and a colleague and I stopped the lesson we had planned and built a class segment on what is and is not okay to say about what somebody is wearing. That segment is Lesson 5.2 in this unit, and it is the reason the unit opens with what clothing communicates instead of with a needle.

The consumer half of this unit I can teach cold. I have a business background, I priced a whole cooking project off a real supermarket circular with my students, and cost per wear is the same move as unit price: divide the money by the thing you actually get. Lesson 5.13 is division, and Lesson 5.12 is a hoodie on a table with four price tags and four vendor types, and those two days are the ones I would show a principal first.

Here is what I am honest about. I had not taught sewing before I built this unit. I observed a colleague's sewing labs and I analyzed one of them for a course, so I have watched a room of twelve-year-olds hold needles, but I had never run it myself. So the hand sewing progression in Topic 5.2 is written step by step, one skill a day, with a demonstration rig for the back row, the six needle rules, and a count-out and count-in routine that runs every single day tools are out. It is written that way so it can be taught from the page, and taught well, by me in my first year of it and by whoever follows me.

The claim the unit makes is small and it is specific. A student who leaves this class can sew a button back on, fix a hem, judge whether a garment is worth the money, and say what a piece of clothing communicates without insulting anybody. Four things. Every one of them is checkable, and every one of them is useful the week after the unit ends.

## 2. Content identified (the concept map, written out)

### 2.1 What the NYS module lists for this unit

The Apparel and Textile Production and Design module has 19 lettered lines across five topic areas. All nineteen are now printed in a lesson's own standards block. Two used to be carried at unit level and both are closed: **ATPD 1. e)** was taught in Lessons 5.1, 5.12, and 5.13 without being printed anywhere, and is now printed in Lesson 5.12, where 1. f) already sat and where the line is assessed on objective 3; **ATPD 4. c) Create a clothing budget** was a real content gap, not a citation problem, and it is now taught in the one-season budget box in Lesson 5.14 Day 2 Step 2, four minutes inside the period. Section 14, items 7 and 8, records both fixes.

Text verbatim from NYS Middle Level CTE FACS Content Module 03, Apparel and Textile Production and Design (nyctecenter.org, June 2018 preliminary release).

| Line                  | Text (verbatim)                                                                                         | Taught in                                                                      |
|-----------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| ATPD 1. Selection a)  | Determine personal needs, wants, and values                                                             | Lessons 5.1, 5.3                                                               |
| ATPD 1. b)            | Identify appropriate clothing for individual roles and activities                                       | Lessons 5.1, 5.2                                                               |
| ATPD 1. c)            | Describe ways personal image and style is projected to others through clothing choices                  | Lessons 5.2, 5.18                                                              |
| ATPD 1. d)            | Inventory personal wardrobe, noting condition and appropriateness of apparel items                      | Lesson 5.13                                                                    |
| ATPD 1. e)            | Select wardrobe items based on personal clothing values and available resources                         | Lessons 5.1, 5.12, 5.13. Printed in Lesson 5.12, where objective 3 assesses it |
| ATPD 1. f)            | Select appropriate clothing for a variety of situations without exceeding financial and other resources | Lesson 5.12                                                                    |
| ATPD 2. Production a) | Identify common equipment and supplies                                                                  | Lessons 5.5, 5.6, 5.10, 5.11                                                   |

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|                                           | used in hand and machine sewing                                                                             |                                                                                                                                                                                   |
| ATPD 2. b)                                | Demonstrate safe use and storage of equipment used in hand and machine sewing                               | Lessons 5.5, 5.7, 5.10, 5.11, 5.16, 5.17                                                                                                                                          |
| ATPD 2. c)                                | Demonstrate basic hand and machine sewing techniques by creating or repairing a textile product             | Lessons 5.6, 5.7, 5.8, 5.9, 5.11, 5.16, 5.17, 5.18. <b>Met in full, hand and machine, since the D19 machine revision; the no-machine fallback note is at the end of Section 7</b> |
| ATPD 2. d)                                | Develop sustainability practices for repurposing apparel items                                              | Lessons 5.14, 5.15, 5.16, 5.17, 5.18                                                                                                                                              |
| ATPD 3. Maintenance a)                    | Identify care procedures which may extend the wear of an apparel item                                       | Lessons 5.3, 5.4, 5.8, 5.13                                                                                                                                                       |
| ATPD 3. b)                                | Use care label information to select appropriate procedures for care of clothing and accessories            | Lesson 5.4                                                                                                                                                                        |
| ATPD 3. c)                                | Identify daily, weekly, and seasonal care of clothing and accessories                                       | Lesson 5.4                                                                                                                                                                        |
| ATPD 4. Consumer Resources and Finance a) | Compare and contrast the cost of constructing an apparel item versus purchasing one or repurposing one      | Lessons 5.9, 5.12, 5.13, 5.15                                                                                                                                                     |
| ATPD 4. b)                                | Compare cost of similar apparel items when obtained from a variety of different types of vendors            | Lessons 5.12, 5.14                                                                                                                                                                |
| ATPD 4. c)                                | Create a clothing budget                                                                                    | Lesson 5.14, the one-season budget box in Day 2 Step 2                                                                                                                            |
| ATPD 4. d)                                | Describe ways that personal decisions regarding apparel and textile choices include a global responsibility | Lesson 5.14                                                                                                                                                                       |
| ATPD 5. Career Pathways a)                | Identify career paths in the                                                                                | Lesson 5.19                                                                                                                                                                       |

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|            | textiles and apparel field                                                                       |             |
| ATPD 5. b) | Investigate a career in clothing or textiles and identify the pathways used to reach that career | Lesson 5.19 |

## 2.2 How the CTE themes come in

- **Sustainability (primary):** the whole back half of the unit runs on it. SUS 1. Resources e) and SUS 3. Energy Conservation c) carry the care and repair days, because a garment that gets washed correctly and a button that gets sewn back on are garments that did not get replaced; SUS 1. f) and SUS 2. Environmental Management e) and j) carry Lessons 5.13 and 5.14, where students price a sustainability decision and read a landfill statistic for its source and year; and the Upcycle Challenge is the Sustainability module's own illustrative activity for Family and Consumer Sciences, which names a recycle, upcycle, or repurposing project and gives t-shirts into shopping bags as the example.
- **Financial and Consumer Literacy:** FCL 4. Buying Goods and Services c), f), g), and h) carry Topic 5.3. Students compare the same hooded sweatshirt across four vendor types, compute cost per wear, run a five-point quality inspection, build a landed cost from a real packing slip, and make a recommendation for a buyer who is not them. FCL 4. h) also sits in Lesson 5.4, because a care label is in the garment by law and that is the reason it is there at all.
- **Problem Solving and Innovation:** PSI 2. Design Process (Proactive) a) runs across the project, one sub-line at a time: defining the problem, the criteria, and the constraints in Lesson 5.15, selecting and prototyping in Lesson 5.16, testing against the criteria in Lesson 5.11 and again in Lesson 5.18, and evaluating their own use of the process in the Lesson 5.18 reflection. PSI 3. Troubleshooting Process (Reactive) a) is the diagnosis step in the Repair Clinic and the stuck protocol on both build days.
- **Health, Safety, and Wellness:** HSW 4. Classroom and Workplace Safety is the thread that is never cut. Five of its nine lines are taught: a) in Lesson 5.3 before the burn demonstration, c) on every day a tool is in a hand, d) in Lesson 5.1 with the real protective gear on the front table, e) in the housekeeping routine that ends every station day, and i) in Lesson 5.11 where eleven days of the count routine is finally scored.
- **Communication and Interpersonal Relationships:** CIR 1. Communication b) and c) are Lesson 5.2, where clothing is treated as nonverbal communication and the class writes its own agreement about what is and is not okay to say. CIR 1. f) and CIR 2. Listening c) are Lesson 5.18, where every student gives a 45-second explanation and writes two stars and a step on somebody else's work.
- **Career and Community Opportunities:** CCO 1. The Work World c) is the workplace dress code in Lesson 5.1. CCO 2. Career Clusters a) and CCO 4. Career Plans a) carry Lesson 5.19, thirteen jobs, three education paths, a "would I like this" self-check, and the five-repair portfolio that a sixteen-year-old can actually show somebody.

## 2.3 Trends, issues, and news

1. **Fast fashion and haul culture.** Students watch people open bags of clothes on a screen. Lesson 5.14 takes that apart with four features (speed, volume, price, thin construction), states the costs that are not on the tag in two sentences each, and says the other side out loud: cheap clothing is real help to a family with little money. The sentence that keeps it working is “I am not telling you where to shop.”
2. **Resale and thrift apps.** Buying and selling used clothing on a phone is normal now. Lesson 5.12 makes the online marketplace one of the four vendor types and builds the landed cost off a real packing slip, shipping and returns included.
3. **Sneaker and clothing resale as a teen business.** Some students already buy to flip. Lesson 5.13’s cost per wear and Lesson 5.12’s quality checks are exactly the tools a reseller uses, and Lesson 5.19 names the two jobs on the list that are commonly run as a small business.
4. **Repair culture and visible mending.** Mending as something to show rather than hide is a current thing, and Lesson 5.9’s patch station offers the visible mend in bright floss as one of two real options.
5. **Dress codes and school policy debates.** Students argue about this already. Lesson 5.1 analyzes a real workplace dress code rule for what it protects and what it costs the worker, and the school’s own code gets quoted exactly, not paraphrased.
6. **Microplastics from synthetic fabric.** Synthetic fleece sheds fiber in the wash. Lesson 5.3 identifies polyester and nylon by test, and Lesson 5.14 names the microfiber shed as one of the three costs that are not on the tag, with its source and year on the slide.

## 2.4 What students this age find interesting

1. Making a real object with their own hands that somebody else will use.
2. Being trusted with a sharp tool, on a routine, with a job title attached (the Station Manager runs the count).
3. Money, and finding out that the best made sweatshirt on the table was also the cheapest.
4. Their own clothes, their own style, and the question of who gets to comment on it.
5. Destroying something on purpose in front of the class: the pull test on two seams, the sweater ruined in a hot wash, the three snips of fiber over a pie pan.

## 2.5 Links to other subjects

- **Math:** cost per wear is division with a unit written on the answer (5.13, 5.14). Measuring to the half inch and the quarter inch, seam allowance, finished size against cut size, and perimeter are the whole of Lessons 5.10 and 5.16. Stitches per inch is counting a rate. The landed cost in Lesson 5.12 is a total with shipping and tax. The repair-or-replace sheet in Lesson 5.9 is a break-even comparison.
- **Science:** fibers are materials, and Lesson 5.3 tests six of them for absorbency, stretch, and wicking, then burns three (cotton burns and leaves crushable ash, wool curls and smells like hair, polyester melts and leaves a hard bead) to make one safety point about melting near heat. Polyester and nylon are polymers, which the science teacher

may already have named. Lesson 5.4's stain station is applied chemistry: a base for grease, an acid for the juice, a solvent for the ink, one variable per cup.

- **ELA:** the display card and the 45-second explanation in Lesson 5.18 are a short informative text and a short informative presentation with a required structure. The fact table in Lesson 5.14 is nonfiction reading: who published it, what year, what does it count, what does it not count. Two stars and a step is a peer feedback protocol the ELA teacher probably already uses.
- **Social studies:** where clothes are made, who makes them, and what a country's garment industry is worth to it. Lesson 5.14's journey of one sweatshirt ends in a bale sold to a used-clothing dealer for export, which is global trade from the plate side.
- **Art:** color, texture, and pattern are the design vocabulary students already met in Unit 4's Color in the Home work, used again here on fabric. Lesson 5.16 cites National FCS 16.3.3 on the elements and principles of design in constructing and altering apparel, and Lesson 5.18's three display tables are composition.

### 3. Enduring understandings and essential questions

#### Big ideas:

- Clothing communicates whether or not you meant it to, and what it says depends on the setting, the audience, and who is looking.
- What a garment is made of decides how it behaves: how it feels, whether it stretches, whether it dries, how it must be washed, and how it fails.
- Most clothes are thrown away long before they stop working. Four skills, a hem, a seam, a patch, and a button, are the difference between a garment being finished and being fixable.
- The price on the tag is not the cost. The cost is the price divided by the number of times you actually wear it, plus what you cannot see: shipping, returns, water, labor, and where it goes when you are done.
- A tool that can hurt somebody gets a routine, not a warning. The count-out and count-in is how a room of twenty-one people uses needles safely every day.
- Making something takes a plan with measurements on it, and the plan gets better when somebody makes you defend it before you cut.

#### Essential questions:

1. What does what I am wearing say about me, and who gets to decide what it says?
2. How do I know what a piece of clothing is made of, and what does that tell me about how to keep it alive?
3. Is this worth fixing, and how do I know?
4. What does this shirt actually cost me, and what does it cost that is not on the receipt?
5. What can I make out of something somebody else threw away?
6. Who does this for a living, and what would it take for me to be one of them?

## 4. Knowledge and skills

### Knowledge (facts, vocabulary, ideas):

- The four jobs clothing does: protect, fit the activity, fit the setting, say something. Personal protective equipment, dress code, uniform, closed-toe shoe, layer.
- Clothing as nonverbal communication: first impression, personal style, cultural dress, generation, intent against impact.
- Natural fibers (cotton, wool, silk, linen) and their sources; manufactured fibers (polyester, nylon, rayon, spandex); blend; what each one is good and bad at; wicking, absorbent, breathable, static.
- What a burn test shows: cotton burns to crushable ash, wool curls and self-extinguishes, polyester melts to a hard bead, and a fiber that melts is dangerous near heat.
- The five care symbol families (wash, bleach, dry, iron, professional care) and what dots, bars, and X marks mean. Care labels are required by law, which is why the tag is there.
- The laundry sequence from sort to fold, the three care groups, and what ruins clothes: hot water on wool, high heat on spandex, bleach on color, a stain that was never pretreated.
- Every tool in the station kit by name and job, and where each one lives when it is not in a hand. Six needle rules. The six steps of the count-out and count-in.
- Four stitches and what each is for: running stitch (basting, gathering, a channel), backstitch (a seam that carries weight), whipstitch (joining two edges, finishing a raw edge), the button stitch with a shank. Blanket stitch on the no-machine fallback path.
- Seam, seam line, seam allowance, right side and wrong side, hem, patch, visible mending, shank, pattern, mock-up.
- Seven parts of a sewing machine, the four machine safety rules, the machine close-down routine, the broken needle rule, and the machine doctor tag. The threading path, top and bobbin.
- Four vendor types and the cost advantage and risk of each. Five hidden costs of buying online. Four quality checks you can run in a store in under a minute, and the five-point inspection: fiber content, seam type, stitches per inch, reinforcement at stress points, buttons and closures.
- Cost per wear as a unit rate. Upcycle against recycle against repurpose. Fast fashion in four features. Where a donated garment actually goes: sorting center, resale floor, bale for export, wiping rags, shredded fiber, landfill.
- Thirteen textile and apparel careers, three education paths, the two that are commonly small businesses, and what a five-repair portfolio is.

### Skills (what students will be able to do):

- Select clothing and footwear for five situations and justify each by weather, activity, and setting, and evaluate one real workplace dress code rule for what it protects and what it costs.

- Analyze how the same shirt reads differently in three settings, and write and defend one line of a class agreement about commenting on what people wear.
- Test six unlabeled swatches by feel, stretch, and water drop, record what each does, and choose the best fiber for a stated use using two performance characteristics from their own data.
- Read a care label, sort a basket of real garments into three care groups, state the laundry sequence in order, and name the step where a ruined garment went wrong.
- Compare three treatments on one stain and compute the cost of the treatment against the cost of replacing the garment.
- Name ten hand sewing tools and their storage places, state the six needle rules, and run a count-out and count-in with a matching number as Station Manager.
- Thread a needle, tie a knot that holds, and sew a running stitch, a backstitch, a whipstitch, and a sewn-on button with a shank that holds a firm tug, then judge each against even length, straight line, and no puckering.
- Diagnose three kinds of garment damage and complete at least two of a hem, a seam repair, and a patch with the correct stitch for each.
- Decide repair or replace with the five-step decision model and defend it with the numbers.
- Measure, mark, and cut from a paper pattern to within a quarter inch, pin, construct a drawstring bag on the machine (or a felt pouch by hand), and evaluate it against a five-point checklist.
- Identify seven machine parts and four machine rules, thread the top and bobbin verified by a buddy, drive a straight line, a curve, and a pivoted square on paper, and sew a straight machine seam reverse-locked at both ends.
- Assemble a six-part stitch sampler, hand and machine, and self-score it before it is scored.
- Run four quality checks on a garment in hand, including counting stitches per inch, and recommend one item for a stated buyer against a second option using price, quality, and one hidden cost.
- Compute cost per wear, rank four garments from most to least likely to last with a reason for each rank, and build a three-option comparison for a real clothing decision of their own.
- Read a statistic for its source, its year, what it counts, and what it does not.
- Write a design brief with a used starting material, a named user, two stitches, an order of work, the one thing most likely to go wrong, and a reduced-scope version, then plan it with four or more measurements, prototype the hardest part, and revise from what the mock-up showed.
- Build the planned item by hand in two periods using the stuck protocol, self-check it against a seven-line finishing checklist, and fix one problem they found themselves.
- Write a four-field display card, deliver a four-part 45-second explanation pointing at the item, give two-stars-and-a-step feedback that names a specific feature, and evaluate their own use of the design process.

- Place thirteen careers on three education paths, complete the fit self-check honestly, and start a five-repair portfolio with this project as entry one.

## 5. Unit objectives

One per topic, upper half of Bloom's:

- **Topic 5.1:** Students will analyze what clothing does and says by matching clothing to five situations, comparing one garment across three settings, and testing six fibers, then evaluate a care label and a stain treatment against the cost of replacing the garment.
- **Topic 5.2:** Students will demonstrate and evaluate four hand sewing techniques and a full station safety routine by producing a six-part stitch sampler, two completed repairs, and one finished hand-sewn item judged against a stitch quality checklist.
- **Topic 5.3:** Students will evaluate clothing purchases by comparing the same item across four vendor types, computing cost per wear, inspecting four garments for construction quality, and recommending one option for a named buyer with a defense that uses price, quality, and one cost that is not on the tag.
- **Topic 5.4:** Students will design and create a useful object from a used textile by writing a design brief against four constraints, prototyping the hardest part, building it by hand in two periods with two of the four taught stitches, and evaluating both the finished item and their own use of the design process.
- **Topic 5.5:** Students will evaluate thirteen textile and apparel careers against their own interests and the training each requires, and justify one first step a middle school student could actually take.

## 6. Topics, sequence, and timeline

| Topic                           | Days                | Lessons                                                                                                                                                                                                 | Activity lines (verb first)                                                                                                                                                                                          | Formal check at end of topic                                                                                                       |
|---------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>5.1 What We Wear and Why</b> | 5 (days 116 to 120) | 5.1 Clothing for Roles, Activities, Weather, and Safety (1); 5.2 What Clothing Communicates (1); 5.3 Fibers: Natural and Manufactured (1); 5.4 Care Labels and Laundry and the Fibers and Care Quiz (2) | list the four jobs clothing does; walk eight real pieces of protective gear and name what each one protects; match clothing and footwear to five situations and justify each; evaluate one workplace dress code rule | Assessment 05.1 - Fibers and Care Quiz.md, 12 items, 24 points, in the last 15 minutes of Lesson 5.4 Day 2 (starting at minute 25) |

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|  |  |  | <p>for what it protects and what it costs; analyze one shirt shown in three settings; write and defend one line of the class agreement; sort eight fibers into natural and manufactured; test six unlabeled swatches by feel, stretch, and water drop; watch the burn demonstration (or the video and wrinkle test) and record what each fiber did; select the best swatch for a stated use with two characteristics as evidence; read the five care symbol families; sort a basket of real garments into three care groups; state the laundry sequence from sort to fold; compare three treatments on one stain; calculate replacement</p> |  |
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|                                           |                      |                                                                                                                                                                                                                                                                                                                                     | cost against treatment cost                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                      |
| <b>5.2 Hand and Machine Sewing Skills</b> | 11 (days 121 to 131) | 5.5 Sewing Tools and Safety (1); 5.6 Threading, Knotting, and the Running Stitch (1); 5.7 Backstitch and Whipstitch (1); 5.8 Sewing On a Button (1); 5.9 Repair Clinic: Hem, Seam, Patch (1); 5.10 Meet the Machine: Parts, Safety, Threading, and Paper Drills (3); 5.11 Build the Drawstring Bag and the Stitch Sampler Check (3) | identify ten tools by name, job, and storage place; state the six needle rules; run the count-out and count-in with a matching number; evaluate three unsafe station scenarios; cut thread to hand-to-elbow length, thread a needle, and tie a knot that holds; sew a running stitch on a printed line; predict which of two seams fails the pull test and say why; sew two inches of backstitch and two inches of whipstitch; select the right stitch for three jobs; sew a two-hole and a four-hole button with a shank that holds a tug; calculate what one repair saved against replacing the garment; diagnose three | Assessment 05.2 - Stitch Sampler Performance Check.md, seven rows, 28 points, in the last 20 minutes of Lesson 5.11 Day 3 (starting at minute 20). The constructed item is scored separately on the 20-point Constructed Item rubric inside Lesson 05.11, live across its three days |

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|  |  |  | <p>kinds of damage;<br/>complete two of three repairs (hem, seam, patch); decide repair or replace with the five-step model and the numbers;<br/>measure, mark, and cut two pieces from a paper pattern;<br/>pin, backstitch the seams, and add a closure;<br/>evaluate the finished item against the five-point checklist;<br/>identify seven machine parts and four machine rules;<br/>thread top and bobbin verified by a buddy;<br/>drive line, curve, and pivoted square on paper;<br/>construct the drawstring bag on the machine (or the felt pouch by hand): cut, pin, seam, channel, turn, cord;<br/>assemble and self-score the six-part sampler card</p> |  |
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| <b>5.3 Smart Apparel Consumers</b> | 4 (days 132 to 135) | 5.12 Comparing the Same Item Across Stores, Secondhand, and Online (1); 5.13 Cost Per Wear and Quality Checks (1); 5.14 Fast Fashion and Where Clothes Go and the Consumer Comparison Task (2) | list one advantage and one risk for four vendor types; run four quality checks on a garment in hand, including counting stitches per inch on a seam; build the landed cost from a real packing slip; recommend one of the four inspected garments for a stated buyer and defend it in writing; compute cost per wear and write the unit; inspect four garments on five points and rank them from most to least likely to last; describe fast fashion in four features and name two costs that are not on the tag; order the seven cards of one sweatshirt's journey and mark the endings; identify a statistic's | The Consumer Comparison Task, 10 points, scored on the printed six-line checklist in Handout 05.14, in the last 20 minutes of Lesson 5.14 Day 2 |
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|                                      |                     |                                                                                                                                   | <p>publisher, year, and what it counts and does not count; create a three-option comparison for a real clothing decision and recommend one with a reason</p>                                                                                                                                                                                                                                                                                         |                                                                                                                       |
| <b>5.4 Upcycle Challenge project</b> | 6 (days 136 to 141) | 5.15 Upcycle Challenge Design Brief (1); 5.16 Plan and Prototype (1); 5.17 Make (2); 5.18 Fashion Show Gallery and Reflection (2) | <p>distinguish upcycling from recycling with a clothing example of each; analyze three paths from the idea menu against the four constraints; create a design brief naming the used material, the user, two stitches, the order of work, the likeliest failure, and a reduced-scope version; label a plan sketch with four or more measurements and a seam allowance; construct a paper mock-up of the hardest part and revise the plan from it;</p> | Rubric 05 - Upcycle Challenge.md, five criteria, 20 points, from the folder collected at the end of Lesson 5.18 Day 2 |

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|                    |                     |                                                                                   | <p>evaluate a partner's plan against the four constraints; defend the plan at the approval table before anything is cut; cut, mark, pin, and stitch the first full seam; apply the three-step stuck protocol before asking for help; finish the item and test the closure with real contents; self-check seven finishing lines and fix one problem found alone; write a four-field display card; deliver the 45-second explanation four times; write two stars and a step on two other items; evaluate their own use of the design process</p> |                                                                                                                  |
| <b>5.5 Careers</b> | 2 (days 142 to 143) | 5.19 Careers in Textiles, Apparel, and Design, and the Unit 5 Review and Test (2) | <p>match thirteen careers to what the person does all day; sort them onto three education paths and mark the two commonly</p>                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Assessment 05.3 - Unit 5 Test.md, 20 items, 50 points plus 2 bonus, in the last 20 minutes of Lesson 5.19</p> |

|  |  |  |                                                                                                                                                                                                                      |                                                                             |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|  |  |  | run as small businesses; complete the eight-question fit self-check honestly; start the five-repair portfolio with this project as entry one; play Make It, Mend It, Grab It as a station review; take the unit test | Day 2. Topic 5.5 has no separate check; it is assessed inside the unit test |
|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|

Day counts:  $5 + 11 + 4 + 6 + 2 = 28$ . The day-by-day placement is in Calendar 05.md.

**Flex lessons (drop first when time runs short), in this order:**

1. **Lesson 5.10 Day 3** (fold the straight-line paper drill into the first ten minutes of Lesson 5.11 Day 1 with no thread; drop the curve and the square). Saves 1 day and costs the pivot practice.
2. **Lesson 5.13 folded into 5.12** (cost per wear becomes the buyer choice's math step; the four-garment inspection ranking is dropped; the Consumer Comparison Task still checks the skill). Saves 1 day and costs the best math day in the topic.
3. **Lesson 5.4 Day 2** (run the stain station as one teacher demonstration inside Day 1, drop the laundry sequence practice to a slide, and give the Fibers and Care Quiz in the last 15 minutes of Day 1). Saves 1 day and costs the stain chemistry.
4. **Lesson 5.1 folded into Lesson 5.2** (keep the PPE walk as a ten-minute segment and the dress code discussion as five minutes of the agreement day; ATPD 1. a) and 1. b) are still taught). Saves 1 day, and costs the most of the four, because the PPE table is the cheapest safety teaching in the unit.

**Why the hand block stays even with a room full of machines (D19):** Lessons 5.6 to 5.9 are the four day hand foundation, threading, running stitch, backstitch, button, and hem, and they run before any student touches a machine. They are the diagnostic that tells me who needs extra eyes at a machine, they are the machine-is-down alternative that keeps a jam from stalling a lesson, and they are the reduced-dexterity alternative D14 requires. A student who can control a hand needle has already learned the only thing a machine cannot teach: patience with fabric.

**Lessons that must not be cut, and one rule that is not negotiable: Lesson 5.5 and the count-out and count-in routine may never be cut, shortened, or moved.** Nobody in this room touches a needle before the six needle rules are taught and the count routine has been practiced for real, and a student who misses day 121 gets the rules and one practice count at my table before their first sewing day, per Assessment 05.2. Also never cut:

Lesson 5.16 (nothing gets cut without an initialed plan, so without this day the build fails), both days of Lesson 5.17 (the build itself), and Lessons 5.6 and 5.7 (the running stitch and the backstitch are what every later day is built on).

**The 24-day version (for the 30-week schedule, which the scope and sequence sets at 24 days for this unit):** drop all four flex items above. Topic 5.1 becomes 3 days (5.1 and 5.2 combined, 5.3, 5.4 in one day with the quiz in the last 15 minutes), Topic 5.2 becomes 10 days (5.10 in two days: parts, rules, and threading in one, paper drills in one), Topic 5.3 becomes 3 days (5.12 and 5.13 combined), Topic 5.4 stays at 6, Topic 5.5 stays at 2. Total 24. The project keeps all six days, which is the point: it is the thing a principal will ask about, and it is the only place students build something.

**The 15-day version (for the one-semester or every-other-day schedule, which the scope and sequence sets at 15 days):**

| Day | Lesson               | What survives                                                                                                                                                                                |
|-----|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | 5.1 and 5.2 combined | The four jobs of clothing, the PPE walk, the same shirt in three settings, and the class agreement                                                                                           |
| 2   | 5.3                  | Natural and manufactured fibers, the three test stations, the burn demonstration or its video version                                                                                        |
| 3   | 5.4 in one day       | The five care symbol families, the basket sort, the laundry sequence, the stain station as a teacher demonstration, and a 10-item version of the Fibers and Care Quiz in the last 15 minutes |
| 4   | 5.5                  | Tools, the six needle rules, the count routine practiced for real. Never cut                                                                                                                 |
| 5   | 5.6                  | Thread, knot, running stitch                                                                                                                                                                 |
| 6   | 5.7                  | Backstitch and whipstitch, with the pull test                                                                                                                                                |
| 7   | 5.8                  | The button and the shank, with the cost math                                                                                                                                                 |
| 8   | 5.9                  | The Repair Clinic and the repair-or-replace decision                                                                                                                                         |
| 9   | 5.10 in one day      | The seven parts, the four                                                                                                                                                                    |

|    |                                    |                                                                                                                                                                                                                                                                   |
|----|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                    | rules, the close-down routine, and the threading drill; paper drills only if machines allow                                                                                                                                                                       |
| 10 | 5.11 in one day                    | The bag cut, pinned, and seamed on pre-threaded machines (or the felt pouch by hand), with the Stitch Sampler Performance Check in the last 20 minutes (six pieces become five: running stitch, backstitch, whipstitch, button, machine seam from the bag itself) |
| 11 | 5.13                               | Cost per wear and the five-point inspection on four garments                                                                                                                                                                                                      |
| 12 | 5.14 Day 1, with the task attached | Fast fashion in four features, the journey cards, and the Consumer Comparison Task cut to two options and 15 minutes                                                                                                                                              |
| 13 | 5.15 and 5.16 combined             | The four constraints, the idea menu, the brief, the plan sheet, and the approval conversation. Nothing is cut without initials                                                                                                                                    |
| 14 | 5.17 Day 1                         | Cut, mark, pin, first seam, stuck protocol                                                                                                                                                                                                                        |
| 15 | 5.17 Day 2, with the card          | Finish, the seven-line checklist, the display card, and a ten-minute gallery with the 45 seconds given once to one partner                                                                                                                                        |

What the 15-day version loses, said plainly: the channel and the turn (the bag closes with a ribbon tie instead), the repair box on the sampler, Lesson 5.12 (the four-vendor comparison), Lesson 5.18 Day 2 (the full gallery, the vote, and the written reflection), and Lesson 5.19 (careers). **ATPD 5. a) and 5. b) then have no home in this unit and move to Unit 7**, where the thirteen career cards become part of the career research; say so in the syllabus. The unit test becomes a 10-item version given as the do now and first fifteen minutes of Unit 6's first day, or it is dropped and the project rubric carries the grade.

**Every-other-day version (a year of 90 meetings):** the 15-day plan, with one addition and one warning. The build gets a third meeting, for 16 meetings, because pinned work sitting in a bag for two days comes back unpinned. And the scrap bin and the donation drive have to open three weeks earlier than day 116 on that calendar, because the bin fills at the rate the class meets.

## 7. Standards block

**NYS module lines taught (every line, by code, with the lesson that teaches it)**

Text verbatim from NYS Middle Level CTE FACS Content Module 03, Apparel and Textile Production and Design (nyctecenter.org, June 2018 preliminary release).

- NYS ATPD 1. Selection a) Determine personal needs, wants, and values : Lessons 5.1, 5.3
- NYS ATPD 1. Selection b) Identify appropriate clothing for individual roles and activities : Lessons 5.1, 5.2
- NYS ATPD 1. Selection c) Describe ways personal image and style is projected to others through clothing choices : Lessons 5.2, 5.18
- NYS ATPD 1. Selection d) Inventory personal wardrobe, noting condition and appropriateness of apparel items : Lesson 5.13
- NYS ATPD 1. Selection e) Select wardrobe items based on personal clothing values and available resources : Lessons 5.1, 5.12, 5.13 (printed in Lesson 5.12)
- NYS ATPD 1. Selection f) Select appropriate clothing for a variety of situations without exceeding financial and other resources : Lesson 5.12
- NYS ATPD 2. Production a) Identify common equipment and supplies used in hand and machine sewing : Lessons 5.5, 5.6, 5.10, 5.11
- NYS ATPD 2. Production b) Demonstrate safe use and storage of equipment used in hand and machine sewing : Lessons 5.5, 5.7, 5.10, 5.11, 5.16, 5.17
- NYS ATPD 2. Production c) Demonstrate basic hand and machine sewing techniques by creating or repairing a textile product : Lessons 5.6, 5.7, 5.8, 5.9, 5.11, 5.16, 5.17, 5.18 (see the coverage note at the end of this section)
- NYS ATPD 2. Production d) Develop sustainability practices for repurposing apparel items : Lessons 5.14, 5.15, 5.16, 5.17, 5.18
- NYS ATPD 3. Maintenance a) Identify care procedures which may extend the wear of an apparel item : Lessons 5.3, 5.4, 5.8, 5.13
- NYS ATPD 3. Maintenance b) Use care label information to select appropriate procedures for care of clothing and accessories : Lesson 5.4
- NYS ATPD 3. Maintenance c) Identify daily, weekly, and seasonal care of clothing and accessories : Lesson 5.4
- NYS ATPD 4. Consumer Resources and Finance a) Compare and contrast the cost of constructing an apparel item versus purchasing one or repurposing one : Lessons 5.9, 5.12, 5.13, 5.15
- NYS ATPD 4. Consumer Resources and Finance b) Compare cost of similar apparel items when obtained from a variety of different types of vendors : Lessons 5.12, 5.14

- NYS ATPD 4. Consumer Resources and Finance c) Create a clothing budget : Lesson 5.14 (the one-season budget box, Day 2 Step 2)
- NYS ATPD 4. Consumer Resources and Finance d) Describe ways that personal decisions regarding apparel and textile choices include a global responsibility : Lesson 5.14
- NYS ATPD 5. Career Pathways a) Identify career paths in the textiles and apparel field : Lesson 5.19
- NYS ATPD 5. Career Pathways b) Investigate a career in clothing or textiles and identify the pathways used to reach that career : Lesson 5.19

**CTE theme lines used (text verbatim from the theme modules, exactly as the lessons cite them)**

### **Sustainability (primary):**

- CTE SUS 1. Resources e) Investigate practices that promote stewardship of environmental resources : Lessons 5.4, 5.15
- CTE SUS 1. Resources f) Research the personal, environmental, and financial costs and benefits of sustainability-conscious decisions to individuals, families, schools, workplaces and communities : Lessons 5.13, 5.14
- CTE SUS 2. Environmental Management e) Explain factors to consider when assessing the environmental impact of purchasing decisions for everyday goods : Lessons 5.3, 5.14
- CTE SUS 2. Environmental Management j) Examine the impacts of landfills, including toxins, leachate, and greenhouse gases, on the global community : Lesson 5.14
- CTE SUS 3. Energy Conservation c) Understand and explain how individuals can limit the use of nonrenewable resources by reducing, reusing, and recycling : Lesson 5.8
- CTE SUS 4. Careers Related to Sustainability b) Analyze career paths within the natural resources, environmental management, and energy conservation fields : Lesson 5.19

### **Health, Safety, and Wellness:**

- CTE HSW 4. Classroom and Workplace Safety a) List and describe the purpose of safety rules specific to classroom and workplace settings : Lesson 5.3
- CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment : Lessons 5.5, 5.6, 5.7, 5.8, 5.10, 5.11, 5.16, 5.17
- CTE HSW 4. Classroom and Workplace Safety d) Wear appropriate protective clothing and gear when using tools, machines, and equipment : Lesson 5.1
- CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace : Lessons 5.5, 5.17
- CTE HSW 4. Classroom and Workplace Safety i) Demonstrate development of workplace safety skills through practice of these skills in a variety of classroom applications : Lessons 5.10, 5.11

### **Financial and Consumer Literacy:**

- CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses : Lesson 5.14 (the one-season budget box)
- CTE FCL 4. Buying Goods and Services f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services : Lessons 5.9, 5.12, 5.13
- CTE FCL 4. Buying Goods and Services g) Compare local, national, and global resources for obtaining goods and services : Lesson 5.12
- CTE FCL 4. Buying Goods and Services h) Analyze consumer rights, responsibilities, and protections provided by local, state, federal, and global laws and policies : Lesson 5.4

### **Problem Solving and Innovation:**

- CTE PSI 2. Design Process (Proactive) a) Implement a formal design process to solve a given problem by a. Defining the problem being addressed; b. Defining criteria that must be met through the finished design; c. Defining constraints that must be adhered to; d. Brainstorming and examining possible solutions : Lesson 5.15
- CTE PSI 2. Design Process (Proactive) a) e. Selecting the best solution for evaluation; f. Developing and constructing a prototype or model of the selected design : Lesson 5.16
- CTE PSI 2. Design Process (Proactive) a) b. Defining criteria that must be met through the finished design; f. Developing and constructing a prototype or model of the selected design; g. Testing and evaluating the prototype and model against the design criteria and constraints : Lesson 5.11
- CTE PSI 2. Design Process (Proactive) a) g. Testing and evaluating the prototype and model against the design criteria and constraints; i. Evaluating their use of the design process and how it impacted their final solutions : Lesson 5.18
- CTE PSI 2. Design Process (Proactive) b) Demonstrate personal development of design skills through practice of these skills in a variety of classroom applications : Lessons 5.6, 5.7
- CTE PSI 3. Troubleshooting Process (Reactive) a) Implement a formal troubleshooting process to solve a given problem by c. Testing and evaluating to isolate the problem; d. Correcting the problem by implementing changes or repairs : Lesson 5.9
- CTE PSI 3. Troubleshooting Process (Reactive) a) Implement a formal troubleshooting process to solve a given problem by a. Defining the problem being addressed; c. Testing and evaluating to isolate the problem; d. Correcting the problem by implementing changes or repairs : Lesson 5.17

### **Communication and Interpersonal Relationships:**

- CTE CIR 1. Communication b) Describe types of communication (e.g., oral, verbal, nonverbal, digital) : Lesson 5.2
- CTE CIR 1. Communication c) Examine communication styles and their effects on relationships : Lesson 5.2
- CTE CIR 1. Communication f) Demonstrate personal development of communication skills through practice of these skills in a variety of classroom applications : Lesson 5.18

- CTE CIR 2. Listening c) Show understanding of active listening and feedback techniques : Lesson 5.18

### **Career and Community Opportunities:**

- CTE CCO 1. The Work World c) Compare the similarities and differences among home, school, community, and work environments : Lesson 5.1
- CTE CCO 2. Career Clusters a) Identify and use career resources to obtain information about careers and employment trends : Lesson 5.19
- CTE CCO 4. Career Plans a) Examine potential career choices to determine knowledge, skills, and abilities associated with each : Lesson 5.19

### **NYS FACS Learning Standards (Intermediate), and the honest answer to the Standard 2 question**

**The state's own Standards Addressed block for the Apparel and Textile Production and Design module lists NYS FACS Learning Standard 2 only.** For sixteen of the nineteen ATPD lines that is fine, because Standard 2's third performance indicator is written about clothing. For three lessons it is not fine, and this unit handles that the way the crosswalk recommends in Part E.3, option (b): cite Standard 3 first where Standard 3 is the honest fit, cite Standard 2 second, and say in one sentence on the lesson page that the state's block lists only Standard 2.

The three lessons that cite Standard 3 first are **Lesson 5.12, Lesson 5.13, and Lesson 5.19.** In 5.12 and 5.13 the content is consumer finance and division, which is ATPD 4. Consumer Resources and Finance; in 5.19 it is careers, which is ATPD 5. Career Pathways. No Standard 2 performance indicator fits either one. Crosswalk Part E.3 names exactly these cases: "ATPD 4. Consumer Resources and Finance (a to c) fits Standard 3 PI B and PI C (budgeting) better than Standard 2 PI C" and "HDR 7. and ATPD 5. Career Pathways fit Standard 3 PI E." Lesson 5.14 cites Standard 2 first and notes that Standard 3 PI B also fits its Day 2 task, because the lesson's own content is global responsibility rather than money.

**Standard 2: A Safe and Healthy Environment.** Students will acquire the knowledge and ability necessary to create and maintain a safe and healthy environment. Key Idea: Students will know the basic principles of home and community safety. They can demonstrate the skills necessary to maintain their homes and workplaces in a safe and comfortable condition. They can provide a safe and nurturing environment for themselves and others.

- PI C: Students use age-appropriate techniques to select and maintain clothing. (Lessons 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 5.9, 5.10, 5.11, 5.14, 5.15, 5.16, 5.17, 5.18, and Project 05. This is the unit's default citation and it is on seventeen of the nineteen lesson pages.)
- PI B: Students know the basics of managing a safe and healthy home. (Not cited on a lesson page in this unit. The laundry content in Lesson 5.4 is the closest fit and PI C covers it better, because PI C names maintaining clothing directly.)

**Standard 3: Resource Management.** Students will understand and be able to manage their personal and community resources. Key Idea: Students will understand and be able to

manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.

- PI B: Students understand the resources available to them, make informed decisions about the use of those resources, and know some ways to expand resources. (Lesson 5.12, cited first; also noted as a fit on Lesson 5.14 Day 2)
- PI C: Students are able to budget their time and money. (Lesson 5.13, cited first)
- PI E: Students identify their own abilities and interests as possible guides to career choice. (Lesson 5.19, cited first)

The PI letters are NYLearns codes; the NYSED print document lists the indicators as bullets, and the wording above comes from NYLearns because the 1996 NYSED PDF is blocked to automated fetch.

### CDOS (Intermediate)

- **CDOS 1 Career Development:** Students will be knowledgeable about the world of work, explore career options, and relate personal skills, aptitudes, and abilities to future career decisions. (Lesson 5.19)
- **CDOS 2 Integrated Learning:** Students will demonstrate how academic knowledge and skills are applied in the workplace and other settings. (Lessons 5.11, 5.17; measuring, marking, and building to a plan is math and procedure doing real work)
- **CDOS 3a Universal Foundation Skills:** Students will demonstrate mastery of the foundation skills and competencies essential for success in the workplace. Sub skills used in this unit:
  - **Basic Skills:** Students listen to and read the ideas of others and analyze what they hear and read; acquire and use information from a variety of sources; and apply a combination of mathematical operations to solve problems in oral or written form. (Lessons 5.3, 5.4, 5.6, 5.7, 5.13)
  - **Thinking Skills:** Students evaluate facts, solve advanced problems, and make decisions by applying logic and reasoning skills. (Lessons 5.9, 5.14, 5.15, 5.16)
  - **Personal Qualities:** Students demonstrate an understanding of the relationship between individuals and society and interact with others in a positive manner. (Lessons 5.1, 5.5, 5.8, 5.11, 5.17)
  - **Interpersonal Skills:** Students demonstrate the ability to work with others, present facts that support arguments, listen to dissenting points of view, and reach a shared decision. (Lessons 5.2, 5.18)
  - **Technology:** Students select and use appropriate technology to complete a task. (Lessons 5.10, 5.11)
  - **Managing Information:** Students select and communicate information in an appropriate format (e.g., oral, written, graphic, pictorial, multimedia). (Lesson 5.18)
  - **Managing Resources:** Students understand the material, human, and financial resources needed to accomplish tasks and activities. (Lesson 5.12)

Standard 3b (Career Majors) exists only at Commencement. A grade 6 to 8 course cites 1, 2, and 3a only, which is what all twelve state modules do. The ATPD module PDF prints “Standards 3a” instead of “Standard 3a”; cite “CDOS 3a” (crosswalk D.4).

**National FCS 3.0 (LEAD FCS Education, 2018 to 2028; text verbatim from the 3.0 compilation as quoted in 02 Standards/STANDARDS SOURCES - web research.md)**

The ATPD module’s own Standards Addressed block names Area 16 at the content-standard level only (16.1, 16.2, 16.4) plus 2.1.4. Numbers marked (block) are printed in the ATPD block; numbers marked (ext) are confirmed verbatim in the 3.0 text but not printed in the module block. All twenty competencies below were checked against section 4.16 (and 4.1, 4.2) of the web research file.

### **Area 16, Textiles, Fashion, and Apparel:**

- 16.2 Evaluate textiles, fashion, and apparel products and materials and their use in diverse settings. (block, content standard) (Lesson 5.1, cited at the standard level because no Area 16 competency covers clothing for roles and activities)
- 16.1.1 Explain what people in textiles, fashion, and apparel careers do. (ext; 16.1 in block) (Lesson 5.19)
- 16.1.2 Examine openings for jobs and for starting a business. (ext; 16.1 in block) (Lesson 5.19)
- 16.1.3 Sum up the education and training that textile, fashion, and apparel career paths require. (ext; 16.1 in block) (Lesson 5.19)
- 16.2.1 Use correct terms to identify, compare, and examine the common generic textile fibers and fabrics. (ext; 16.2 in block) (Lesson 5.3)
- 16.2.2 Judge how textile fibers and fabrics perform. (ext; 16.2 in block) (Lesson 5.3)
- 16.2.3 Examine textile laws, standards, and labeling in a global economy. (ext) (Lesson 5.4)
- 16.2.4 Examine how a textile’s components affect its design, construction, care, use, upkeep, and disposal or recycling. (ext) (Lesson 5.3)
- 16.2.5 Show the right way to care for textile products and to dispose of or recycle them, keeping local and global needs in mind. (ext) (Lessons 5.4, 5.15, 5.18, and the grade 8 stretch CARE field on the display card)
- 16.2.6 Judge fibers and fabrics on sustainability. (ext) (Lesson 5.14)
- 16.2.7 Judge the construction quality and fit of textiles, fashion, and apparel. (ext) (Lessons 5.7, 5.11, 5.13)
- 16.3.1 Explain how fiber, fabric, texture, pattern, and finish change how a garment looks. (ext) (Lesson 5.2)
- 16.3.3 Use the elements and principles of design when designing, making, or altering textiles and apparel. (ext) (Lesson 5.16)
- 16.4.1 Work skillfully with traditional and newer equipment, tools, and supplies to construct, alter, repair, and recycle textiles and apparel. (ext; 16.4 in block) (Lessons 5.5, 5.6, 5.10, 5.11, 5.17)

- 16.4.3 Use the right industry products and materials to clean, press, and finish textiles and apparel. (ext) (Lesson 5.4)
- 16.4.5 Show the basic skills for making, altering, repairing, and recycling textiles and apparel. (ext; 16.4 in block) (Lessons 5.6, 5.7, 5.8, 5.9, 5.10, 5.11, 5.16, 5.17, 5.18)
- 16.5.2 Examine what it costs to construct, manufacture, distribute, alter, repair, or recycle textiles, fashion, and apparel. (ext) (Lessons 5.9, 5.12, 5.13)

**Competencies from other areas the lessons cite:**

- 1.2.1 Examine possible careers to find what each one requires and offers in knowledge, skills, attitudes, and openings. (ext) (Lesson 5.19)
- 1.2.7 Examine what keeps school, work, and community settings safe and healthy. (ext) (Lesson 5.5)
- 2.1.4 Use consumer skills when buying and caring for clothing. (block) (Lessons 5.1, 5.8, 5.12)
- 2.2.1 Examine the part individuals and families play in current environmental trends and issues. (ext) (Lesson 5.14)
- 2.2.3 Practice habits that conserve, reuse, and recycle so resources and the environment last. (ext) (Lesson 5.15, Project 05, Rubric 05 criterion 1)
- 2.6.2 Use sound money management principles in personal and family finances. (ext) (Lesson 5.14, the one-season budget box)

**Not cited, and why:** no Area 16 competency covers clothing and identity, which is what ATPD 1. c) is about, so Lesson 5.2 cites 16.3.1 for the visual-appearance half and lets the state line carry the identity half. 16.2.6 would also fit ATPD 4. d) at unit level. Credit LEAD FCS Education visibly on the site; it is a condition of the non-commercial terms of use.

**Career Ready Practices (as the ATPD module lists them; wording from the extract's assembled list)**

The ATPD module PDF cites seven of the twelve:

- CRP 1. Act as a responsible and contributing citizen and employee
- CRP 2. Apply appropriate and academic and technical skills
- CRP 3. Attend to personal health and financial well-being
- CRP 5. Consider environmental, social, and economic impacts of decisions
- CRP 6. Demonstrate creativity and innovation
- CRP 11. Use technology to enhance productivity
- CRP 12. Work productively in teams while using cultural global competence

CRP 2 is printed “Apply appropriate and academic and technical skills” in the ATPD PDF, which is the state’s own typo; the Nutrition and Wellness PDF prints the corrected “Apply appropriate academic and technical skills.” The line above is quoted as the ATPD module prints it. No single state PDF prints all twelve practices, so the extract assembles the list from all twelve modules, and Career Ready Practices are a secondary citation on this unit page only. The CCTC source text has not been fetched (crosswalk E.2).

## The coverage note, updated for the machine revision (D19)

**Since the 2026-09-14 revision, ATPD 2. c) is met in full, hand and machine.** The room has machines for most of the class at once (D19), Lesson 5.10 teaches the machine as three days of parts, safety, threading, and paper drills, and Lesson 5.11 constructs a real machine-sewn bag. The hand half of 2. c) is met nine lessons deep, and the machine half is now demonstrated, not diagrammed. **The honest note now runs the other way:** if this program lands in a room with no working machines, the archived hand-sewn build replaces Lessons 5.10 and 5.11, the machine half of 2. c) drops back to taught-from-the-diagram, and the website's Unit 5 page must say so. The machine parts, rules, and threading path are still taught from Handout 05.10 - Machine Parts and Threading Guide.md in that fallback, so 2. a) and 2. b) hold either way.

## 8. Project

**Name:** Upcycle Challenge. Full sheet in Project 05 - Upcycle Challenge.md (handed out in Lesson 5.15 with the rubric); scored on Rubric 05 - Upcycle Challenge.md.

**The scenario:** find a used textile that was headed for the trash, turn it into something a real person will actually use, build it by hand in two class periods with at least two of the four stitches the class already knows, and explain it in 45 seconds to people who were not there while it was made. Individual work at a station of about seven, with a station partner for help and peer review. Six days, days 136 to 141.

**The four constraints, every path, no exceptions:** it starts from a used textile, not new fabric off a roll; it is finished in two build days; it uses at least two of the taught joins, and since D19 a machine seam counts as one of them (fabric glue or hem tape can hold one join, two joins are stitched, by hand or machine); and it is something somebody would actually use, with that person named. The machines are available on both build days for any student whose plan calls for them, with the same rules as Topic 5.2; no plan is required to use one.

**Deliverables (eight, all named in Project 05):** (1) the design brief, with the starting material's size and condition, the finished item, the named user, the two stitches, three steps in order, the likeliest failure, and the reduced-scope version; (2) the plan sheet with four or more measurements, finished size and cut size with the seam allowance, the three marks, the numbered order of work, the mock-up of the tricky part, what the mock-up changed, the peer review, and my initials; (3) the finished item; (4) tool safety and the station routine across all four days tools are out; (5) the four-field display card, name on the back only; (6) the four-part 45-second explanation; (7) the five-prompt reflection; (8) the self-assessment row of Rubric 05, filled before I score.

**Steps by day:** Lesson 5.15 (day 136) upcycle against recycle, the ten paths, the four constraints, pick a material, write the brief, rubric goes out. Lesson 5.16 (day 137) plan sheet, mock-up, the approval conversation at my table (nothing is cut before I initial), peer review. Lesson 5.17 Day 1 (day 138) count-out, cut, mark, pin, first full seam, stuck protocol, milestone check, count-in. Lesson 5.17 Day 2 (day 139) finish, reduced-scope call at minute 10 if behind, the seven-line finishing checklist self-checked and initialed. Lesson 5.18 Day 1 (day 140) ten-minute finishing window, display card, the 45-second planner,

gallery set up on three tables, two timed practice rounds. Lesson 5.18 Day 2 (day 141)  
gallery walk with two feedback slips minimum, explanations in rotation, the three-category vote, the reflection, the folder in.

**Group or individual:** individual, at a station of about seven. The only role is Station Manager, and it rotates every sewing day so three different students run a count.

**Choice built in:** the path (ten examples plus design your own), the material (scrap bin, donation drive, or home with a signed note), the user, which two of the four stitches and where, how much to add (a reinforcement, a second row of stitching, a detail, or a second small item), and how to present (a rotation, one partner, or one-on-one with me).

**ELL and IEP supports:** the picture plan template, where the brief and plan are a labeled drawing with number blanks and are scored on the same criterion 1 descriptors; sentence starters in Turkish, Portuguese, and Spanish, all to be verified by a native speaker; the reduced-scope version written on day 136 so no decision is made under pressure; the reduced-dexterity kit (self-threading needles, large-eye needles with heavier thread, threaders, pre-threaded needles at my table, binder clips instead of pins, a clamp or tape to hold the fabric, felt instead of fraying woven fabric, glue or hem tape on one of the two joins, a larger item with longer stitches), which never appears on a score; pre-measured and pre-marked material on request; the no-presentation option (the card plus a one-on-one 45 seconds, offered privately, scored identically); extended time into the next lesson's first minutes and the day 140 finishing window; and a written alternative for an absence on a build day. Full list in Project 05, Supports.

**How it is presented:** three tables, items and display cards only, no bodies and no modeling. Nobody wears anything in front of the room unless they ask to. Each maker gives the 45 seconds three or four times to whoever is standing at their item. Three named vote categories keep neatness from winning everything: Most Useful, Most Surprising Material, Best Construction. Every item is photographed with its display card, no faces and no names, and that photo set is the Unit 5 page on the website, the hallway display case, and next year's Lesson 5.15 slides at once.

**Full no-machine version:** the project runs entirely by hand, exactly as it did before the D19 revision, because a machine is an option here, never a requirement. What does need a stated alternative is dexterity, not machinery, and the reduced-dexterity kit above is required equipment on both build days, not an accommodation to be found on the day.

## 9. Assessment plan

- **Formative (daily):** the do now every day, a named check for understanding in every lesson (a three-box or four-box clipboard walk at the stations, a tug test on a button, a stitch count read aloud, thumbs on a sample row, a cold call for a statistic's year), an exit card or a one-line share most days, and the station count sheet initialed every day tools are out. All daily work is scored complete, partial, or missing.
- **Formal at the end of each topic:**
  - **Topic 5.1:** Assessment 05.1 - Fibers and Care Quiz.md, 12 items, **24 points**, in the **last 15 minutes of Lesson 5.4 Day 2 (day 120), starting at**

**minute 25.** Matching on fibers, a four-part care symbol item, sorting, two ruined-garment scenarios, and one math item with a calculator allowed. Extended time finishes in the first five minutes of Lesson 5.5, where the do now is a tool sort and nothing is missed.

- **Topic 5.2:** two pieces, and they are scored in different categories. Assessment 05.2 - Stitch Sampler Performance Check.md, seven rows scored 1 to 4, **28 points**, in the **last 20 minutes of Lesson 5.11 Day 3 (day 131), starting at minute 20**, at the table, out loud, about 40 seconds a student, three tables at a time. That is the topic assessment. The **Constructed Item rubric, 20 points, five rows, lives inside Lesson 05.11** and is scored live on the clipboard across days 129 to 131 (rows 1 and 5 on Day 1, rows 2 and 5 on Day 2, rows 3, 4, and 5 on Day 3); it is the labs and projects score for the bag or pouch, and it is what the scope and sequence calls the item rubric.
- **Topic 5.3:** the **Consumer Comparison Task, 10 points**, scored on the printed six-line checklist on Handout 05.14 - Fast Fashion Facts and Consumer Comparison Task.md, in the **last 20 minutes of Lesson 5.14 Day 2 (day 135)**. Three options with price, cost per wear shown as division, one quality check each, one sustainability consideration, and a recommendation with a reason tied to the buyer and the use.
- **Topic 5.4:** Rubric 05 - Upcycle Challenge.md, five criteria at four levels, **20 points**, scored from the folder collected at the end of Lesson 5.18 Day 2 (day 141) and returned with the item at the end of day 143. Criterion 4 is scored across all four days tools are out, from the daily clipboard, not from one observation.
- **Topic 5.5:** no separate check; assessed inside the unit test.
- **Unit test:** Assessment 05.3 - Unit 5 Test.md, 20 items, **50 points plus 2 bonus**, in the **last 20 minutes of Lesson 5.19 Day 2 (day 143)**, after a 15-minute station review game that stops at minute 18 whatever the score is. Matching, short answer, three sorting items, two math items with calculators, one scenario, and one written response. A grade 6 version drops the second prompt of the written response and sets up the cost-per-wear division. One retake after a review, best score counts.
- **Grade weights within the unit** follow the course grading plan: labs and projects 40 percent (the 20-point Constructed Item rubric and the 20-point Rubric 05), topic assessments 25 percent (the Fibers and Care Quiz 24, the Stitch Sampler Performance Check 28, the Consumer Comparison Task 10, and the unit test 50), daily work 20 percent (do nows, handouts, exit cards, the practice cloth and sampler pieces, the project's working papers), participation and safety 15 percent (the count-out and count-in on all twelve sewing days, the machine close-down routine on all six machine days, the housekeeping routine, and the stuck protocol). **A tool safety violation is a 1 on row 7 of the sampler check, which zeroes that week's participation and safety score and starts a conversation, per the grading plan.**
- **Marking period note:** days 116 to 135 are in marking period 3 and days 136 to 143 are in marking period 4. The Fibers and Care Quiz, the Constructed Item rubric, the Stitch Sampler Performance Check, and the Consumer Comparison Task land in MP3. Rubric 05 and the unit test land in MP4. That split matters when the MP3 grade closes

on day 135: the project has not started yet, so MP3's labs and projects weight rests entirely on the Constructed Item rubric.

- **Missed days:** Assessment 05.2 carries the full absence policy and it is unusually detailed because eleven days of skills stack. One or two missed piece days: make the pieces at a side table or a free machine during Lesson 5.11's work blocks or at a lunch period, full credit. Three or more: a reduced sampler of four pieces (backstitch, button, one repair, machine seam), five rows out of 20 scaled to 28, with one full catch-up period. Missed the Lesson 5.10 machine days: the threading license and control boxes are earned at a lunch period before sewing at a machine alone. Missed the Lesson 5.11 build days: the item is finished at a lunch period or in the Topic 5.4 build days and the item rubric is scored on what was possible, scaled to 20. **Missed Lesson 5.5: no student sews until the six needle rules and the count routine have been taught to them, one to one, at my table, in the first five minutes of their first day back.** A missed build day gets the written alternative in Project 05 and the day 140 finishing window. A missed quiz or test is taken during the next do now or in a support period.

## 10. Materials, budget, and room

**Room assumptions:** a FACS room with three station tables seating about seven each, chairs at a height that lets an elbow rest on the table (most dropped needles come from a bad working height), wall space for the six needle rules at every station plus the class agreement from Lesson 5.2 and the finishing checklist during the build, a projector, a document camera if one exists, a sink, and a floor that has been swept before the unit starts so a needle found on day 130 is a needle from this unit. Sewing machines are standard equipment in this unit since the D19 revision: numbered machines for most of the class at once, one or two tested spares, and the hand station. The whole unit still runs without them via the archived fallback (Section 7's coverage note). An iron is optional everywhere except pressing the channel fold in Lesson 5.11, and that fold can be pressed by hand.

**The costing below is built from the budget notes the lesson writers left in Lessons 5.3, 5.4, 5.6, 5.8, 5.10, 5.11, 5.15, 5.16, 5.17, and 5.18, and from the station kit table in Lesson 5.5.** The D19 machine line: spare machine needles (a box of ten universal 80/12), one spare bobbin per machine, a small screwdriver, red card stock for the machine doctor tags, and taped-down power strips run about \$25 one time; machine servicing is the building's line, not this budget's. Every price in the lessons is an estimate at 2027 prices. Most of the textile stock in this unit is free: the scrap bin and the donation drive supply the project, and thrifted garments do the rest at about two dollars each.

| Line                                 | Lessons     | What it buys                                           | Low  | High | One time or every year |
|--------------------------------------|-------------|--------------------------------------------------------|------|------|------------------------|
| Fabric shears, 6 pairs at about \$12 | 5.5 to 5.18 | Two per station, marked FABRIC ONLY, per the kit table | \$72 | \$72 | One time               |

|                                                    |                  |                                                                                                                                                                              |      |      |                                 |
|----------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|---------------------------------|
|                                                    |                  | in Lesson 5.5                                                                                                                                                                |      |      |                                 |
| Paper scissors, 3 at about \$2                     | 5.5, 5.10, 5.16  | One per station kit, for patterns and mock-ups, so the shears stay sharp. Lesson 5.16's mock-up day adds two of the room's own classroom pairs per table, which cost nothing | \$6  | \$6  | One time                        |
| Small snips, 9 at about \$2                        | 5.5 to 5.18      | Three per station, for cutting thread close                                                                                                                                  | \$18 | \$18 | One time                        |
| Seam rippers, 6 at about \$3                       | 5.5 to 5.18      | Two per station, kept where students can reach them                                                                                                                          | \$18 | \$18 | One time                        |
| Tailor's chalk, two packs of 3                     | 5.5, 5.10, 5.16  | Marking cut lines and seam lines                                                                                                                                             | \$6  | \$6  | One time                        |
| Measuring tapes, 6 at about \$3                    | 5.5, 5.10, 5.16  | Two per station                                                                                                                                                              | \$18 | \$18 | One time                        |
| Rulers, 3 at about \$2                             | 5.10, 5.12, 5.16 | Quarter-inch marks, also used for counting stitches per inch                                                                                                                 | \$6  | \$6  | One time                        |
| Hand needles, two packs of 25 assorted sharps plus | 5.5 to 5.18      | 10 per station in the cushion, counted every day                                                                                                                             | \$8  | \$8  | One time, replaced as they bend |

|                                                                                                                     |                        |                                                                                          |      |      |          |
|---------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------|------|------|----------|
| large-eye embroidery                                                                                                |                        |                                                                                          |      |      |          |
| Straight pins, two boxes                                                                                            | 5.5 to 5.18            | 30 per station, counted every day                                                        | \$6  | \$6  | One time |
| Pin cushions, 6 at about \$3                                                                                        | 5.5 to 5.18            | Two per station, the only legal place for a needle not in a hand                         | \$18 | \$18 | One time |
| Needle threaders, two packs of 10                                                                                   | 5.5 to 5.18            | Four per station, out in the open so using one is not asking for help                    | \$6  | \$6  | One time |
| Thimbles, two packs                                                                                                 | 5.5 to 5.18            | Four per station, twelve in all. Check the roster for nickel and latex sensitivity first | \$8  | \$8  | One time |
| Magnet on a stick and a hard plastic sharps jar                                                                     | 5.5 to 5.18            | The floor sweep when a count does not match, and bent needles                            | \$8  | \$8  | One time |
| Reduced-dexterity kit: self-threading (calyx eye) needles, large-eye embroidery needles, one box of 12 binder clips | 5.10, 5.16, 5.17, 5.18 | Required equipment on both build days, not an accommodation found on the day             | \$11 | \$11 | One time |

|                                                                       |             |                                                                                                            |      |      |          |
|-----------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------|------|------|----------|
| Three kit boxes with the layout photo taped inside the lid            | 5.5 to 5.18 | A missing tool is visible from across the room                                                             | \$18 | \$18 | One time |
| Three clipboards                                                      | 5.5 to 5.18 | The station count sheets and every live rubric                                                             | \$3  | \$3  | One time |
| Fiber swatch master sets: one thrift bag of six garments at about \$2 | 5.3, 5.4    | Cuts into about twenty labeled sets, stored in the Unit 5 bin                                              | \$12 | \$12 | One time |
| Laundry sort garments: ten thrifted items with their tags             | 5.4         | Must include wool, spandex, white cotton, colored cotton, an iron label, a dry clean label, jeans, a towel | \$20 | \$20 | One time |
| One sweater ruined on purpose in a hot wash                           | 5.4         | The best two dollars in the unit                                                                           | \$2  | \$2  | One time |
| One thrifted button-down shirt with a button missing                  | 5.8         | The hook, sewn shut in front of the class in ninety seconds                                                | \$2  | \$2  | One time |
| Practice cloth stock: one thrifted flat sheet at about \$4            | 5.6, 5.7    | Cuts into about sixty six-by-nine practice cloths with the lines drawn on                                  | \$4  | \$4  | One time |
| Repair Clinic                                                         | 5.9         | Pants or                                                                                                   | \$12 | \$12 | One time |

|                                                                                         |                           |                                                                                            |              |              |            |
|-----------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------|--------------|--------------|------------|
| practice garments: six thrifted items at about \$2                                      |                           | curtain panels for hems, T-shirts or tote bags for seams, jeans or sweatshirts for patches |              |              |            |
| Demonstration rig: blunt yarn needle, bright yarn, burlap or plastic canvas, foam board | 5.6 to 5.10               | What the back row watches                                                                  | \$10         | \$10         | One time   |
| Clothesline and clips, or a small drying rack                                           | 5.4                       | The stain station's drying                                                                 | \$8          | \$8          | One time   |
| Four sample hooded sweatshirts and four inspection garments                             | 5.12, 5.13                | <b>A non-purchase ask first: see below</b>                                                 | \$0          | \$16         | One time   |
| An iron and a small pressing board on a heatproof mat                                   | 5.10 (optional elsewhere) | Only if the iron stays in this unit at all and the room has none                           | \$0          | \$25         | One time   |
| Printing and laminating the kit layout photos and station labels                        | 5.5                       | Three lids, three tables                                                                   | \$0          | \$4          | One time   |
| <b>One-time subtotal</b>                                                                |                           |                                                                                            | <b>\$300</b> | <b>\$345</b> |            |
| Thread, six                                                                             | 5.6 to 5.18               | Contrast                                                                                   | \$18         | \$18         | Every year |

|                                                                                                 |                 |                                                                               |      |      |            |
|-------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------|------|------|------------|
| spools of all-purpose polyester at about \$3                                                    |                 | colors, because white thread on white cloth teaches nobody anything           |      |      |            |
| Craft felt, about 30 sheets at about 30 cents                                                   | 5.7, 5.10, 5.11 | The felt path, the whipstitch rectangles, the pin cushion path                | \$9  | \$9  | Every year |
| Woven fabric for the drawstring path: one thrifted flat sheet or two pillowcases                | 5.10            | Cuts into about fifteen eight-by-eighteen rectangles                          | \$4  | \$4  | Every year |
| Cotton cord or shoelaces, one spool                                                             | 5.10, 5.17      | Drawstrings                                                                   | \$4  | \$4  | Every year |
| Buttons, one jar of assorted                                                                    | 5.8, 5.10, 5.17 | Free if they are cut off donated garments, which is a five-minute student job | \$6  | \$6  | Every year |
| Stain treatment supplies: dish soap, baking soda, white vinegar, rubbing alcohol, swabs, gloves | 5.4             | Pull the Safety Data Sheets for the alcohol and the vinegar                   | \$12 | \$12 | Every year |
| Patch fabric and embroidery                                                                     | 5.9, 5.11       | Bright floss is the point of a visible                                        | \$8  | \$8  | Every year |

|                                                                                                                                                      |                 |                                                                                    |             |             |                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------|-------------|-------------|-----------------------------------------------------------------------------|
| floss for the visible mend                                                                                                                           |                 | mend                                                                               |             |             |                                                                             |
| Fiberfill for the pin cushion path                                                                                                                   | 5.11 Path B     | A handful each; fabric scraps also work                                            | \$4         | \$4         | Every year                                                                  |
| Fabric glue and iron-on hem tape                                                                                                                     | 5.17, 5.18      | Reduced-dexterity path only, at my table                                           | \$8         | \$8         | Every year                                                                  |
| Card stock, two packs                                                                                                                                | 5.8, 5.11, 5.18 | Portfolio cards, sampler cards, display cards. The copier's heaviest paper is free | \$12        | \$12        | Every year                                                                  |
| Masking tape, index cards, and one labeled zip bag per student                                                                                       | 5.3 to 5.18     | Every piece of everybody's work lives in a named bag                               | \$6         | \$6         | Every year                                                                  |
| <b>Consumables subtotal, per year</b>                                                                                                                |                 |                                                                                    | <b>\$91</b> | <b>\$91</b> |                                                                             |
| Project extras: rice (two pounds), binder clips, fabric-marking chalk, butcher paper or three cloths for the display tables, plus the cord, buttons, | 5.15 to 5.18    | Mock-up paper is free (newspaper, brown bags). The textiles themselves are free    | \$20        | \$39        | Every year, and about \$18 of it overlaps the consumables rows; buy it once |

|                                                      |  |  |  |  |  |
|------------------------------------------------------|--|--|--|--|--|
| glue, hem tape, and card stock already counted above |  |  |  |  |  |
|------------------------------------------------------|--|--|--|--|--|

**First-year total: about \$401 to \$457.** That is the \$300 to \$345 of reusable tools and stock, plus \$91 of consumables, plus the roughly \$10 to \$21 of the project line that is genuinely new spend after the overlap is removed. **Later years: about \$91 to \$100**, which is thread, felt, floss, treatment supplies, card stock, and rice, less whatever the donation drive covers. There is no per-student fee and nothing a family has to send in.

**Class size, and what a fourth station costs.** Every lesson, handout, and assessment in this unit is written for **three stations of about seven, 21 students**, which is the same assumption Unit 1 and Unit 6 make. **A section of 28 runs a fourth station**, and the added one-time cost is **about \$60**: two more pairs of fabric shears at about \$12 (\$24), three more snips (\$6), two more seam rippers (\$6), two more measuring tapes (\$6), two more pin cushions (\$6), one more paper scissors (\$2), two more sticks of chalk (\$2), one more kit box with a layout photo (\$6), and one more clipboard (\$1). Needles, pins, threaders, thimbles, and thread all come out of the packs already costed above. A fourth station also adds about four minutes to the Lesson 5.11 sampler check and takes a second thrifted flat sheet for practice cloths, about \$4. Nothing else in the unit changes.

**Where the textiles come from, which is the reason this unit is cheap:** two free sources, both of which have to be started early. The **room scrap bin** opens on **day 116**, the first day of the unit, with a sign on it, and it has to hold at least twenty usable pieces by day 136. The **class donation drive** bin sits by the door and takes clean, used textiles from anybody; contributing is never required and is never tracked, and everything that arrives goes into the shared scrap bin. A student who brings nothing from home builds the same project on the same rubric and is never asked about it in front of the class.

**The two non-purchase asks:** both are borrowing requests, and if both are refused the high column above covers buying the garments thrifted.

1. **Four sample hooded sweatshirts for Lesson 5.12**, one per vendor type: a thin discount-store fleece, a mall brand, a heavy secondhand, and one that arrived in an online order with its shipping bag and packing slip (the packing slip is the part that matters; black out the name, address, and order number). Ask staff, the lost and found with permission, or a thrift run.
2. **Four inspection garments for Lesson 5.13**, labeled 1 to 4, at least one clearly well made and one clearly not: a heavy cotton sweatshirt with double-needle seams and bar tacks, a thin promotional T-shirt with a cracking graphic, a pair of jeans with a flat felled seam and a shank button, and a fast-fashion blouse with a loose button and a fraying serged seam. Two of these can be the Lesson 5.12 sweatshirts, which is why the four sweatshirts are worth keeping in the bin all year; they come back again in the Unit 5 review game.

### Handouts (in 05 Materials/Unit 05/, as the lessons name them):

- Topic 5.1: Handout 05.01 - Match the Clothing to the Situation.md; Handout 05.02 - What Clothing Says and the Class Agreement.md; Handout 05.03 - Fiber Test Station Record.md; Handout 05.04 - Care Symbol Guide and Laundry Sequence.md; Handout 05.04 - Stain Station Record.md
- Topic 5.2: Handout 05.05 - Tool Kit Guide and Count Sheet.md (the station count sheet, used on all twelve sewing days; the machine close-down routine is printed there and in Handout 05.10, its two canonical places); Handout 05.06 - Practice Cloth Guide.md; Handout 05.07 - Backstitch and Whipstitch Steps.md; Handout 05.08 - Button Steps and Portfolio Card.md; Handout 05.09 - Repair Clinic Station Cards.md; Handout 05.10 - Machine Parts and Threading Guide.md (with the drill sheets and the threading license); Handout 05.11 - Bag and Pouch Pattern Cut Guide and Seam Quality Checklist.md; Handout 05.11 - Sampler Card.md
- Topic 5.3: Handout 05.12 - Price and Spec Sheets and Buyer Cards.md; Handout 05.13 - Cost Per Wear Worksheet and Quality Inspection Checklist.md; Handout 05.14 - Fast Fashion Facts and Consumer Comparison Task.md
- Topic 5.4: Handout 05.15 - Upcycle Design Brief and Idea Menu.md (the permission note for items from home is its last page); Handout 05.16 - Plan Sheet and Peer Review.md; Handout 05.17 - Build Milestones and Stuck Protocol.md; Handout 05.18 - Display Card Explanation Planner and Feedback Slip.md
- Topic 5.5: Handout 05.19 - Textiles and Apparel Careers Cards.md (thirteen cards plus the 42-question review game bank in seven categories)

**Slides:** one outline per lesson, Slides 05.01 through Slides 05.19, in the same folder. The site build renders every outline as an HTML deck.

**No-machine fallback for the whole unit:** if the machines fail the pre-unit test, the archived hand build replaces Lessons 5.10 and 5.11 wholesale, the sampler runs at six rows instead of seven, and the coverage note in Section 7 applies. Decide before day 116 and tell the class on day 121 so nobody spends a week hoping for machines. The reduced-dexterity kit is required equipment either way.

## 11. Health and other-department coordination

**Art.** Two real overlaps. **Color, texture, and pattern** are the same vocabulary the art teacher uses and the same vocabulary Unit 4 already taught in the Color in the Home work; in this unit they land on fabric instead of walls, in Lesson 5.3's swatch handling and Lesson 5.16's design decisions, and Lesson 5.16 cites National FCS 16.3.3 on the elements and principles of design in constructing and altering apparel. **The running stitch is a drawn line**, and the art teacher's line-quality words (even, steady, weight) transfer directly to Lesson 5.6's three criteria. FACS gives art a set of practice cloths and a gallery of twenty-one made objects; art gives FACS its own words for color, texture, and pattern so the two rooms say the same thing, and possibly a hand on the day 140 display setup.

**Science.** Two places. **Fibers are materials science:** Lesson 5.3's six-swatch test is a properties investigation, polyester and nylon are polymers, and the burn demonstration belongs in a fume hood more than in a FACS room. **Lesson 5.4's stain station is applied chemistry:** a base for grease, an acid for the juice, a solvent for the ink, three treatments on one stain with everything else held the same. FACS gives science a ready properties lab, a set of labeled swatches, and a class that has already run a three-treatment comparison; science gives FACS the fume hood, its own words for variable and control, and the microfiber content that Lesson 5.14 names.

**Math.** Lesson 5.13 is division with a unit written on the answer, Lesson 5.12 is a total with shipping and tax, Lesson 5.9 is a break-even comparison, and Lessons 5.10 and 5.16 are measurement to the quarter inch, seam allowance, finished size against cut size, and perimeter. Use the math class's own setup and words: cost per wear as a unit rate, stitches per inch as a rate, the plan sheet as a scale drawing at one square to one inch. FACS gives math four real costing problems with garments on the table and a scale drawing that gets built; math gives FACS students who already divide correctly and, if willing, ten minutes on day 133 to check the cost-per-wear work.

**The school nurse.** Notify the nurse **before day 121, the first day needles come out**, and tell her three things: which twelve days of the calendar have needles and pins in the room, what the count-out and count-in routine is and that no student is dismissed on a count that does not match, and that there is a hard plastic sharps jar in the room for bent needles and broken pins. Ask her two things: whether any student on the roster has a nickel or latex sensitivity, which decides the thimbles and the pins, and what she wants done for a needle stick, in her words, so the answer is the same in her office and in my room. FACS gives the nurse a written routine and twelve dated days; the nurse gives FACS a medical protocol and a roster check. **This is not a courtesy, it is the one conversation in this unit that has to happen before the unit starts.**

**The custodian.** Two asks, both small and both early. The **scrap bin and the donation drive bin** sit in the room from day 116 to day 141 and they need to not be emptied as trash, so tell him what they are and put a sign on both. And the **floor**: sweep the room before day 121 so a needle found later belongs to this unit, and expect thread on the floor for ten teaching days. FACS gives the custodian a room that gets cleaned to a routine by students every sewing day and a class that picks up what it drops; the custodian gives FACS the two bins, a warning before any floor work, and (in Lesson 5.1) the loan of a high-visibility vest and a work glove for the PPE table.

**Health.** The overlap is narrow and it is worth naming: the class agreement in Lesson 5.2 is about what students say to each other about appearance, which touches body image and belonging. Anything a student says about being commented on at home or online is a quiet referral to the counselor, not a class discussion, and the lesson says so. The counselor may want a copy of the agreement; give them one. This unit does not teach body image, dietary guidance, or self-esteem content; the health course owns those.

**Physical education.** The PE teacher already enforces footwear and can co-sign the closed-toe rule in one sentence at the door on day 116, which makes the FACS rule read as a building rule instead of a FACS rule.

**Technology or makerspace, if the building has one.** Their tool rules and their tool count are the same idea as ours. Use the same words in both rooms so a student hears one standard, and ask whether their teacher will open a sewing machine cover for ten minutes on Path A, because a machine is a mechanism with a cam, a linkage, and a feed.

## 12. Connections (in-building and community)

### Inside the building:

- **The art teacher**, for Lessons 5.3, 5.16, and 5.18. **FACS gives:** twenty-one made objects and three display tables the art teacher can point at, a set of labeled fabric swatches for texture and pattern work, and a class that already knows the design vocabulary from Unit 4. **FACS gets:** the department's own words for color, texture, and pattern, a critique protocol that works with twelve-year-olds, and a second adult in the room on the day the gallery goes up.
- **The school nurse**, before day 121 and for all twelve sewing days. **FACS gives:** a written count-out and count-in routine, the twelve dated days when needles are in the room, a sharps jar already in place, and students who announce and pick up anything they drop. **FACS gets:** the nickel and latex roster check before thimbles and pins go out, the building's needle-stick protocol in her own words, and a professional who knows what this room is doing before anything happens rather than after.

### Community:

- **A local tailor or alterations shop**, for Lessons 5.9, 5.16, and 5.19. This is the best partner in the unit and the call is worth making before Lesson 5.9, not after. **FACS gives:** a thank-you from the class, twenty-one families who now know there is an alterations shop in walking distance and what it charges, and a class that understands why hemming a pair of pants costs money. **FACS gets:** the real local price for a hem, a seam, and a patch (which is the number that makes Lesson 5.9's repair-or-replace math land), ten minutes on what the job is actually like all day, and the honest answer to "what would you look at if a sixteen-year-old asked you for work."
- **A thrift store, consignment shop, or clothing pantry**, for Lessons 5.8, 5.12, and 5.14. **FACS gives:** a class of customers who can read a fiber content label and count stitches per inch, a thank-you note, and cut-off buttons and usable scraps sorted out of what they cannot sell. **FACS gets:** the answer to "what makes you price a donated sweatshirt at six dollars instead of two," a reject bin the class can cut up for the Repair Clinic and the scrap bin (which is most of this unit's material budget), and the real story of what arrives and what has to be thrown away.
- **A quilt guild, a sewing circle, or a senior center sewing group**, for Lessons 5.6, 5.7, and the build days. **FACS gives:** a room of beginners who genuinely want to be taught, a public thank-you, and photographs of what the class made from what people gave. **FACS gets:** extra pairs of hands on the hardest day of the unit, which is threading day,

plus donated fabric, thread, and buttons, and sometimes two borrowed machines and a volunteer, which turns Path B into a hybrid where the class hand sews and rotates through two machines for the paper-line practice.

- **A BOCES or district CTE fashion, apparel, or design program**, for Lesson 5.19. **FACS gives:** the program a group of eighth graders who have already built something, repaired a garment, inspected four for quality, and started a five-repair portfolio, which is the closest thing to a feeder that a middle school FACS class can be. **FACS gets:** the real program name and its entry requirements for slide 12, one visitor or one recorded answer about what the program does, and a concrete next step to put in front of a student who says “I like clothes.”

### 13. Vocabulary for translation

See Vocabulary 05.md: 66 Tier 3 terms across the five topics with a one-line student definition written for a grade 6 to 7 reader and columns for Turkish, Portuguese, and Spanish, to be checked by a native speaker, a bilingual colleague, or the ENL teacher. Tool and stitch names are the ones most likely to differ by country, and several have no single everyday word in every language, so the English term stays on every card; a student will see the English word on a real care label, a real pattern, and a real tool. The file also carries sentence starters in all four languages for four things: asking for help at a sewing station, the count-out and count-in routine, giving feedback on a classmate’s item, and the 45-second explanation at the gallery. Each lesson names its own Tier 3 words and its handout carries a word bank pulled from the table. The vocabulary list with translations is allowed as a word bank on the Unit 5 Test.